MRIDA OS

by Sketchitup Solutions

PRODUCT REQUIREMENTS DOCUMENT

Fertilizer Trading Firms Vertical

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1. Executive Summary

Mrida OS is Sketchitup Solutions' flagship operating system for agriculture and allied businesses, built — as the tagline puts it — to take an anchor client 'from your shop floor to your bigger vision.' This document specifies the Fertilizer Trading Firms sub-vertical: a purpose-built operating layer for retailers/distributors of fertilizer, pesticides, seeds, and agri-inputs, covering the full flow from supplier procurement through godown inventory, counter sales, farmer credit, delivery, and owner-level business intelligence.

Fertilizer trading in India carries constraints a generic ERP does not natively handle: statutory licensing under the Fertilizer Control Order, Aadhaar-authenticated Point-of-Sale billing for subsidised fertilizer under the government's DBT/iFMS framework, a regulatory cap on how much subsidised fertilizer a single buyer can purchase in a month, and a heavy day-to-day dependence on Tally for statutory accounting that any new system must complement rather than fight. Mrida OS — Fertilizer Trading is designed to be the connected operations and owner-visibility layer sitting alongside Tally: Tally continues to record what happened; Mrida OS shows the owner what is happening now and what needs to happen next.

> **WHY THIS MATTERS**
>
> Every workflow in this PRD is written to be directly buildable by the Sketchitup dev team without external consulting input — each has explicit actors, step-by-step flow, numbered functional requirements, data entities, and business rules, consistent with the documentation standard used for OwnerOS and the Mrida OS — Seed Firms PRD.

2. Product Vision & Strategic Context

2.1 Strategic Fit within Sketchitup

- Mrida OS is the Agri Tech domain flagship, sitting under the Sketchitup Solutions LLP master entity, with IP held centrally and licensed down to the Agri Tech sub-LLP per the existing three-layer legal architecture.
- Fertilizer Trading is a second Mrida OS sub-vertical alongside Seed Firms, extending the platform from seed production/certification into agri-input retail and distribution — the shop-floor end of the same farmer-facing value chain.
- Positioning is explicit and client-validated: 'We're not replacing Tally. We're eliminating fragmentation.' Mrida OS owns operations and owner visibility; Tally continues to own statutory books of account.
- Mirrors the anchor-client incubation model used for OwnerOS (GuruOm Industries) and the Seed Firms vertical: co-develop inside one fertilizer trading client, harden the product, then replicate across that client's dealer/peer network.

2.2 Product Vision

To become the single operating system for fertilizer and agri-input trading businesses — connecting supplier purchase, godown inventory, counter sales, farmer credit, delivery, and compliance on one real-time platform — replacing scattered registers, WhatsApp coordination, and manual reconciliation with connected, owner-visible workflows, without disturbing the accounting system already in place.

2.3 Design Principles

- Complementary to Tally, not competitive — Mrida OS is the operations + visibility layer; Tally remains system of record for statutory accounting (Section 6.10).
- Compliance-by-design — Fertilizer Control Order licensing, PoS/DBT subsidy sales, and statutory stock registers are built into the sales and inventory workflow, not bolted on.
- Batch- and expiry-aware — every inventory record carries batch/lot and expiry visibility, since pesticides and several fertilizer grades are shelf-life sensitive.

- Owner-first visibility — the owner dashboard (sales, stock value, outstanding, alerts) is a first-class citizen of the product, not an afterthought reporting module.
- Configurable, not custom-coded — products, price lists, schemes, and godowns are master-data driven so the same product serves multiple fertilizer trading firms.

3. Objectives & Success Metrics

Objective	Success Metric / Target
Eliminate fragmented, register/WhatsApp-based tracking	100% of purchase, stock, and sales transactions captured in-system within 2 months of go-live
Improve real-time stock visibility	Warehouse/godown stock variance versus system reduced to < 2%
Reduce credit/outstanding risk	Dealer/farmer outstanding ageing beyond 60 days reduced by a defined target within 2 seasons
Eliminate expiry-driven inventory loss	Zero stock sold or written off past expiry without an alert having fired at the configured lead time
Ensure subsidy/compliance accuracy	100% of subsidised fertilizer sales captured with valid PoS/Aadhaar authentication and zero missed monthly-cap violations
Give the owner real-time visibility	Owner dashboard (sales, stock value, outstanding, alerts) usable anytime/anywhere without depending on staff for a manual update
Avoid duplicate bookkeeping	Zero manual re-entry of sales/purchase data into Tally — fully reconciled via sync (Section 6.10)

4. User Personas & Stakeholders

Persona	Role in Workflow	Primary Device
Owner / Proprietor	Views business dashboard, approves credit limits and large purchases, overall visibility	Web + Mobile
Purchase Manager	Supplier management, purchase orders, GRN, purchase invoice booking	Web
Godown / Store Manager	Stock receipt, batch/expiry tracking, stock transfer, physical reconciliation	Web + Mobile
Counter Sales Staff	Billing, PoS/DBT subsidy transactions, sales invoice generation	Web/POS + Mobile
Accountant	Tally-side bookkeeping; consumes synced vouchers/ledgers from Mrida OS	Tally (Desktop)
Delivery / Dispatch Staff	Vehicle assignment, delivery challan, proof of delivery capture	Mobile
Field Sales Officer	Farmer/dealer visits, order booking, scheme communication	Mobile
Customer / Farmer	Purchases fertilizer/agri-inputs, holds credit account, subsidy beneficiary	In-person / SMS
Sub-Dealer / B2B Customer	Bulk buyer with running credit account and periodic settlement	In-person / Portal

Persona	Role in Workflow	Primary Device
Supplier (Manufacturer/Distributor)	Supplies stock against purchase orders; source of purchase pricing	External

5. Scope Definition

5.1 In Scope (Fertilizer Trading Vertical — Phase 1 & Phase 2)

- Supplier/vendor master and purchase management
- Purchase & procurement: purchase orders, goods receipt, purchase invoice booking
- Inventory & godown management: multi-godown, batch/lot, expiry tracking, stock transfer
- Sales & billing: counter sales, GST invoicing, PoS/DBT subsidy billing for regulated fertilizer
- Customer & farmer management: KYC, credit vs. cash accounts, purchase history
- Credit & payments: credit limits, outstanding ledger, collections, ageing
- Delivery & dispatch: vehicle assignment, delivery challan, proof of delivery
- Regulatory & subsidy compliance: FCO license tracking, PoS/Aadhaar-authenticated sales, statutory stock register, monthly purchase-cap enforcement
- Business intelligence: owner dashboard (sales, stock value, outstanding, alerts, top products) and operational reports
- Tally integration: bidirectional sync of ledgers, stock items, and sales/purchase vouchers
- Mobile app for godown staff, delivery staff, field sales officers, and the owner

5.2 Out of Scope (Phase 1 — Candidate for Later Phases)

- E-commerce storefront for direct-to-farmer online ordering
- Manufacturing/blending operations (this vertical covers trading/distribution, not fertilizer production)
- Direct integration with the government iFMS/mFMS central portal beyond PoS-device-level compliance (Phase 1 assumes the certified PoS device/vendor handles central reporting; Mrida OS mirrors the transaction for internal records)
- Multi-currency/export billing
- Payment gateway-based online collection from farmers (assumed cash/bank/UPI manual reconciliation in Phase 1)

6. Detailed Functional Requirements by Workflow

Workflows 6.1–6.7 follow the exact end-to-end business flow validated with the anchor client: Supplier Management → Purchase & Procurement → Inventory & Godown → Sales & Billing → Customer & Farmer → Credit & Payments → Delivery & Dispatch, followed by Business Intelligence, Regulatory Compliance, Tally Integration, Master Data, and the Mobile App. Each follows a consistent structure: Objective, Actors, Process Flow, Functional Requirements, Data Entities, and Business Rules/Edge Cases. FR-IDs are prefixed by module for traceability into the dev backlog.

6.1 Supplier Management

Objective: Maintain the master record of fertilizer/pesticide/seed/agri-input suppliers (manufacturers and distributors) and their commercial terms, since every purchase and landed-cost calculation is keyed off supplier data.

Actors

- Purchase Manager, Owner (large-supplier approval)

Process Flow

1. Register supplier with KYC (firm name, GSTIN, address, bank details) and product categories supplied.
2. Capture supplier-wise price lists and applicable schemes/discounts per product.
3. Record supplier credit terms (credit period, credit limit extended to the firm).
4. Track supplier performance: fill rate, delivery lead time, quality/rejection history.
5. Flag suppliers whose FCO manufacturing/marketing license or GST registration is expiring.

Functional Requirements

- **FR-SUP-01** System shall maintain a Supplier Master with GSTIN, address, bank details, and product categories supplied.
- **FR-SUP-02** System shall maintain supplier-wise price lists with effective-date versioning so historical purchase prices remain traceable.
- **FR-SUP-03** System shall record supplier credit terms (credit period in days, credit limit) used to flag overdue supplier payments.
- **FR-SUP-04** System shall track supplier performance metrics (on-time delivery %, rejection %) computed from Purchase & Procurement records.
- **FR-SUP-05** System shall store supplier statutory documents (GST certificate, manufacturing/marketing license) with expiry reminders.

Data Entities

- Supplier Master, Supplier Price List, Supplier Document, Supplier Performance Record

6.2 Purchase & Procurement

Objective: Convert stock requirements into purchase orders, capture goods receipt against them with batch/expiry detail, and book the purchase invoice — creating the first traceability point for every unit of stock entering the business.

Actors

- Purchase Manager, Godown Manager (receipt confirmation), Owner (approval above threshold)

Process Flow

6. Raise a Purchase Order against a supplier, referencing current stock levels and reorder points from Inventory.

7. Route PO for owner approval where order value exceeds a configurable threshold.
8. Receive goods against the PO: record Goods Receipt Note (GRN) with quantity, batch/lot number, manufacturing date, and expiry date.
9. Flag quantity or quality discrepancies between PO and GRN for follow-up with the supplier.
10. Book the purchase invoice against the GRN, capturing GST input tax details.
11. Update stock ledger and trigger sync of the purchase voucher to Tally (Section 6.10).

Functional Requirements

- **FR-PUR-01** System shall support Purchase Order creation referencing Supplier Master and current stock/reorder levels.
- **FR-PUR-02** System shall support a PO approval workflow with configurable value thresholds requiring owner sign-off.
- **FR-PUR-03** System shall capture Goods Receipt Note (GRN) with quantity, batch/lot number, manufacturing date, and expiry date per line item.
- **FR-PUR-04** System shall flag and log quantity/quality variance between PO and GRN for supplier follow-up.
- **FR-PUR-05** System shall support partial receipt against a single PO (multiple GRNs per order).
- **FR-PUR-06** System shall book the purchase invoice with GST input tax capture and update the stock ledger on confirmation.
- **FR-PUR-07** System shall generate a purchase voucher payload for Tally sync on invoice booking (see FR-TLY-02).

Data Entities

- Purchase Order, GRN, Purchase Invoice, Batch/Lot Record

> **BUSINESS RULE**
>
> Batch/lot number and expiry date are mandatory at GRN for any product flagged as shelf-life sensitive in the Product Master (pesticides, several fertilizer grades, bio-inputs) — this is the anchor for every downstream expiry alert and statutory stock register entry.

6.3 Inventory & Godown Management

Objective: Maintain real-time, batch-wise, godown-wise stock with low-stock and expiring-stock visibility — directly powering the owner dashboard's Key Alerts panel (Low Stock Alerts, Expiring Stock).

Actors

- Godown/Store Manager, Purchase Manager, Sales Staff (stock check), Owner (alerts)

Process Flow

12. Receive stock into godown inventory from GRN, updating batch-wise, product-wise quantity on hand.
13. Configure minimum stock/reorder level per product; system raises a Low Stock Alert when breached.
14. Compute days-to-expiry per batch; system raises an Expiring Stock Alert at a configurable lead time.
15. Support stock transfer between godowns/branches with in-transit visibility.
16. Support periodic physical stock count/reconciliation with variance logging.
17. Reserve/allocate stock against confirmed sales orders before billing (soft allocation).

Functional Requirements

- **FR-INV-01** System shall maintain real-time stock by product, batch, and godown/branch location.
- **FR-INV-02** System shall raise a Low Stock Alert when on-hand quantity falls below the configured reorder level per product/godown.

- **FR-INV-03** System shall raise an Expiring Stock Alert per batch at a configurable lead time before expiry (e.g., 60/30/15 days).
- **FR-INV-04** System shall support inter-godown stock transfer with in-transit status and receiving confirmation.
- **FR-INV-05** System shall support cycle-count and full physical stock reconciliation with variance reporting.
- **FR-INV-06** System shall support FEFO (first-expiry-first-out) picking logic for billing allocation by default, configurable per product.
- **FR-INV-07** System shall compute live Stock Value (quantity × last purchase/weighted-average cost) feeding the owner dashboard.
- **FR-INV-08** System shall block billing of a batch past its expiry date without an explicit, audited manual override.

Data Entities

- Godown Master, Stock Ledger (Product × Batch × Godown), Reorder Level, Stock Transfer, Cycle Count

> **BUSINESS RULE**
> Low Stock Alerts, Expiring Stock Alerts, and Stock Value are not report-time calculations — they must be maintained as live, continuously updated figures so the owner dashboard (FR-BI-01) always reflects current state without a manual refresh/report-run step.

6.4 Sales & Billing

Objective: Generate GST-compliant sales bills at the counter or via field sales, correctly branching between normal retail billing and Aadhaar-authenticated PoS billing for government-subsidised fertilizer grades.

Actors

- Counter Sales Staff, Field Sales Officer, Customer/Farmer, Godown Manager (stock deduction)

Process Flow

18. Create a sales bill by selecting customer (or walk-in), products, and quantities.
19. System determines whether any line item is a subsidised fertilizer grade requiring PoS/Aadhaar billing.
20. For subsidised items: authenticate buyer via the certified PoS device (Aadhaar/Voter ID/KCC) before completing the sale (see FR-CMP-02).
21. For non-subsidised items: bill normally with applicable GST rate and any active scheme/discount.
22. Apply credit-limit check if the customer is billing on account rather than cash (see FR-CRD-02).
23. Allocate stock (FEFO) from the relevant godown/batch and print/share the invoice.
24. Trigger sales voucher sync to Tally and update the customer's outstanding ledger where sold on credit.

Functional Requirements

- **FR-SAL-01** System shall generate GST-compliant sales invoices with product-wise HSN/tax detail.
- **FR-SAL-02** System shall auto-detect subsidised fertilizer line items and route them through the PoS/Aadhaar authentication step before invoice completion.
- **FR-SAL-03** System shall support both cash and credit (on-account) billing, with credit billing subject to the credit-limit check (FR-CRD-02).
- **FR-SAL-04** System shall apply active scheme/discount rules automatically at billing (volume discount, seasonal scheme).
- **FR-SAL-05** System shall allocate stock FEFO by default and decrement the stock ledger on invoice confirmation.
- **FR-SAL-06** System shall support sales returns with reversal of stock and, where applicable, of subsidy-linked transaction records.
- **FR-SAL-07** System shall generate a sales voucher payload for Tally sync on invoice confirmation (see FR-TLY-02).

- **FR-SAL-08** System shall print/share an MRP-compliant invoice showing the government-notified MRP for regulated fertilizer grades where applicable.

Data Entities

- Sales Invoice, Sales Invoice Line, PoS Transaction Reference, Scheme Master, Sales Return

6.5 Customer & Farmer Management

Objective: Maintain a KYC-complete profile of every customer — walk-in farmer, regular credit customer, or sub-dealer — with the identity data needed for subsidy authentication and the purchase history needed for credit decisions and cross-sell.

Actors

- Counter Sales Staff, Field Sales Officer, Farmer/Customer, Sub-Dealer

Process Flow

25. Register customer/farmer with basic KYC (name, mobile, village, Aadhaar/Voter ID reference for subsidy billing where applicable).
26. Classify customer type: Walk-in/Cash, Regular Credit Customer, Sub-Dealer/B2B.
27. Capture landholding detail where available (supports subsidy quantity reasonableness checks).
28. Maintain running purchase history per customer across seasons for cross-sell and credit review.
29. Flag customers approaching or exceeding the regulatory monthly subsidised-purchase cap (see FR-CMP-03).

Functional Requirements

- **FR-CUS-01** System shall maintain a Customer Master with type (Walk-in / Credit / Sub-Dealer), mobile number, and village/address.
- **FR-CUS-02** System shall optionally capture Aadhaar/Voter ID/KCC reference per customer to speed up repeat PoS subsidy billing, stored per applicable data-protection requirements.
- **FR-CUS-03** System shall maintain full purchase history per customer, viewable by Sales Staff and Owner.
- **FR-CUS-04** System shall support customer segmentation (by village/territory, by product preference) for targeted scheme communication (Section 6.9).
- **FR-CUS-05** System shall link a Sub-Dealer/B2B customer to its own downstream credit and outstanding ledger, distinct from retail farmer accounts.

Data Entities

- Customer Master, Customer Type, Purchase History, Landholding Reference

6.6 Credit & Payments

Objective: Control how much credit each customer/sub-dealer can carry, track the running outstanding ledger, and record collections — directly powering the owner dashboard's Outstanding figure and Pending Payments alert.

Actors

- Owner, Counter Sales Staff, Accountant, Customer/Sub-Dealer

Process Flow

30. Assign a credit limit and credit period to a customer/sub-dealer (default cash-only for new walk-in customers).
31. At billing, check outstanding + new invoice value against the credit limit; block or require owner override if exceeded.
32. Record payments received (cash/bank/UPI/cheque) against customer outstanding, oldest-invoice-first by default.
33. Compute outstanding ageing (0–30 / 31–60 / 61–90 / 90+ days) per customer and in aggregate.
34. Trigger payment reminders (SMS/WhatsApp) at configurable ageing thresholds.
35. Sync payment receipts to Tally as receipt vouchers (Section 6.10).

Functional Requirements

- **FR-CRD-01** System shall maintain a Credit Limit and Credit Period per customer/sub-dealer, editable by the Owner.

- **FR-CRD-02** System shall block or flag-for-override any credit sale that would take a customer's outstanding beyond their credit limit.
- **FR-CRD-03** System shall maintain a running Outstanding Ledger per customer, updated on every credit sale, return, and payment.
- **FR-CRD-04** System shall compute ageing buckets (0–30/31–60/61–90/90+ days) per customer and provide a consolidated ageing report.
- **FR-CRD-05** System shall support partial payments and apply them oldest-invoice-first by default, with manual override to a specific invoice.
- **FR-CRD-06** System shall trigger SMS/WhatsApp payment reminders at configurable ageing thresholds.
- **FR-CRD-07** System shall generate a receipt voucher for Tally sync on every payment recorded (see FR-TLY-02).

Data Entities

- Credit Limit, Outstanding Ledger, Payment Receipt, Ageing Report

> **BUSINESS RULE**
> The Outstanding figure on the owner dashboard (FR-BI-01) must be a live sum of the Outstanding Ledger across all customers, not a periodically-refreshed report — credit risk visibility is only useful to the owner if it is current.

6.7 Delivery & Dispatch

Objective: Manage physical delivery of billed stock to the customer's location where the sale is not a walk-in counter pickup, with proof of delivery closing the loop back to the sales invoice.

Actors

- Delivery/Dispatch Staff, Godown Manager, Customer/Farmer

Process Flow

36. Generate a delivery challan against a confirmed sales invoice requiring delivery (not counter pickup).
37. Assign vehicle and delivery staff, and schedule delivery date.
38. Delivery staff confirms physical dispatch quantity matches the invoice/challan before leaving the godown.
39. Capture proof of delivery at destination (customer signature/photo/OTP) via mobile app.
40. Flag and escalate any delivery discrepancy (short delivery, damaged goods) for resolution.

Functional Requirements

- **FR-DEL-01** System shall generate a delivery challan linked to a confirmed sales invoice, listing product/batch/quantity to be delivered.
- **FR-DEL-02** System shall support vehicle and delivery-staff assignment with scheduled delivery date.
- **FR-DEL-03** Mobile app shall allow delivery staff to capture proof of delivery (signature, photo, or OTP) at the point of delivery.
- **FR-DEL-04** System shall flag delivery discrepancies (quantity/damage) against the originating invoice for resolution tracking.
- **FR-DEL-05** System shall provide a delivery status view (Pending / Dispatched / Delivered / Discrepancy) visible to Sales and Owner.

Data Entities

- Delivery Challan, Vehicle/Transport Master, Proof of Delivery, Delivery Discrepancy Log

6.8 Regulatory & Subsidy Compliance

Objective: Keep the firm compliant with the Fertilizer Control Order (FCO) licensing regime and the government's Direct Benefit Transfer (DBT) framework for subsidised fertilizer, under which subsidy is paid to manufacturers only against authenticated retail sales recorded through Aadhaar-enabled PoS devices via the Integrated Fertilizer Management System (iFMS).

Actors

- Owner, Counter Sales Staff, Certified PoS Device (external), State Agriculture/Fertilizer Department (external)

Process Flow

41. Maintain the firm's fertilizer dealer/retailer license(s) under FCO with validity and renewal tracking.
42. For every subsidised fertilizer sale, route the transaction through the shop's certified PoS device for Aadhaar/Voter ID/KCC-based buyer authentication before the sale is finalized.
43. Mirror the PoS-authenticated transaction reference (device transaction ID, buyer identifier hash, quantity) against the Mrida OS sales invoice for internal reconciliation.
44. Check cumulative subsidised-fertilizer quantity purchased by the buyer in the current month against the regulatory cap before allowing the sale to proceed on the PoS device.
45. Maintain the statutory stock register (opening stock, receipts, sales, closing stock per product) in the format required for inspection by the licensing authority.
46. Generate periodic compliance reports (stock register extract, subsidy sales summary) for the owner and, where required, for submission to the department.

Functional Requirements

- **FR-CMP-01** System shall maintain FCO dealer/retailer license records with validity dates and renewal reminders at a configurable lead time.
- **FR-CMP-02** System shall flag every sales line item belonging to a subsidised fertilizer grade and require a linked PoS transaction reference before the invoice can be finalized.
- **FR-CMP-03** System shall track cumulative subsidised-fertilizer quantity per buyer per calendar month and warn Sales Staff when a new sale would breach the regulatory cap (50 bags/buyer/month nationally; 60 bags/buyer/season in Jammu & Kashmir, or the then-current notified limits).
- **FR-CMP-04** System shall maintain a statutory stock register per product (opening stock, receipts, sales, adjustments, closing stock) exportable in the format required for inspection.
- **FR-CMP-05** System shall retain PoS transaction references and buyer authentication evidence for the statutory record-retention period.
- **FR-CMP-06** System shall generate a subsidy sales summary report reconciling Mrida OS sales records against PoS device transaction logs, flagging any mismatch for investigation.
- **FR-CMP-07** System shall display the government-notified MRP for regulated fertilizer grades on the product master and enforce it as a ceiling price at billing.

Data Entities

- FCO License Record, PoS Transaction Reference, Monthly Purchase Cap Tracker, Statutory Stock Register, MRP Master

> **BUSINESS RULE**
>
> Mrida OS does not replace the certified PoS device or the government's iFMS/mFMS reporting — the licensed PoS hardware/software remains the system of record for subsidy claims. Mrida OS's role is to mirror, reconcile, and surface that data for internal stock, credit, and compliance visibility (Phase 1 scope boundary — see Section 5.2).

6.9 Business Intelligence & Dashboards

Objective: Give the owner the real-time, anytime/anywhere business view that is the core promise of Mrida OS — directly reflecting the validated dashboard design: headline figures, key alerts, and top products.

Actors

- Owner, Sales Manager, Purchase Manager

Functional Requirements

- **FR-BI-01** System shall provide an Owner Dashboard showing, at minimum: Total Sales (with vs.-last-month trend), Stock Value (with trend), and Outstanding (with customer count), matching the validated product design.
- **FR-BI-02** System shall provide a Key Alerts panel showing live counts for Low Stock Alerts, Expiring Stock, and Pending Payments, each drilling down to the underlying records.
- **FR-BI-03** System shall provide a Top Products view ranked by sales volume/value over a selectable period (default: current month).
- **FR-BI-04** System shall provide role-specific views: a Purchase dashboard (open POs, pending GRNs, supplier dues) and a Sales dashboard (dealer/customer-wise sales, credit exposure).
- **FR-BI-05** System shall support scheduled report export (PDF/Excel) and email/WhatsApp delivery to configured recipients.
- **FR-BI-06** System shall provide drill-down from any dashboard figure down to the underlying transaction record.
- **FR-BI-07** Owner Dashboard shall be accessible on both web and mobile with equivalent core figures, consistent with the product's ‘anytime, anywhere’ positioning.

Data Entities

- Dashboard Metric Snapshot, Alert Definition, Top Products Ranking

6.10 Tally Integration & Accounting

Objective: Ensure Mrida OS transactions flow into the client's existing Tally system without duplicate manual entry, positioning Mrida OS as the operations and visibility layer while Tally remains the statutory book of account — per the validated positioning ‘Tally records what happened, Mrida OS helps you understand what is happening and what needs to happen next.’

Actors

- Accountant, Owner, Tally (external system)

Process Flow

47. Map Mrida OS master data (products → Tally Stock Items, customers/suppliers → Tally Ledgers) to existing Tally masters during onboarding.
48. On each confirmed transaction (purchase invoice, sales invoice, payment/receipt) in Mrida OS, generate the equivalent Tally voucher payload.
49. Push the voucher to Tally via XML-over-HTTP against the Tally instance's configured port (Tally's standard integration interface, run in server mode).
50. Log sync status per voucher (Pending / Synced / Failed) with retry and manual-resync capability.
51. Optionally pull ledger balances and stock summaries back from Tally via its ODBC interface for reconciliation/reporting cross-checks.
52. Surface any sync failures to the Accountant/Owner for manual resolution before month-end close.

Functional Requirements

- **FR-TLY-01** System shall support a one-time master data mapping between Mrida OS Product/Customer/Supplier records and existing Tally Stock Item/Ledger records.
- **FR-TLY-02** System shall generate and push Tally-compatible voucher payloads (Sales, Purchase, Receipt, Payment) via XML-over-HTTP on transaction confirmation.
- **FR-TLY-03** System shall maintain a per-voucher sync status (Pending / Synced / Failed) with automatic retry and manual resync action.
- **FR-TLY-04** System shall support read-only pull of ledger balances and stock summaries from Tally via its ODBC interface for cross-reconciliation.
- **FR-TLY-05** System shall alert the Accountant/Owner of any unsynced or failed vouchers before a configurable cutoff (e.g., end of day, month-end).
- **FR-TLY-06** System shall never overwrite or delete data directly inside Tally; sync is additive/one-directional for vouchers (Mrida OS → Tally) except for the read-only balance pull in FR-TLY-04.
- **FR-TLY-07** System shall log every sync attempt (timestamp, payload, response) for audit and troubleshooting.

Data Entities

- Tally Master Mapping, Voucher Sync Queue, Sync Log

> **BUSINESS RULE**
>
> Tally Prime exposes no native REST API or webhooks — integration is via XML-over-HTTP (voucher push, master read) and ODBC (read-only reporting), both requiring a running, network-reachable Tally instance. Sync is therefore poll/push-based, not real-time event-driven; the dev team should design for eventual consistency and build the sync-status view (FR-TLY-03) as a first-class screen, not an afterthought.

6.11 Master Data Management

Objective: Centralize the configuration data every other module depends on, so Mrida OS can be configured per fertilizer trading firm without code changes.

Functional Requirements

- **FR-MDM-01** System shall provide a central admin console for Product/SKU, Price List, Scheme, Godown/Location, and Vehicle masters.
- **FR-MDM-02** Product Master shall support categorization (Fertilizer — Subsidised / Fertilizer — Non-subsidised / Pesticide / Seed / Other Agri-Input) driving PoS-routing and shelf-life-tracking behavior.
- **FR-MDM-03** System shall support role-based access control (RBAC) with configurable permission sets per persona (see Section 4).
- **FR-MDM-04** System shall support multi-branch/multi-godown configuration for firms operating more than one outlet.
- **FR-MDM-05** System shall maintain a full change-audit log for all master data edits (who, what, when, before/after value).

Data Entities

- Product/SKU Master, Price List, Scheme Master, Godown Master, Vehicle Master, RBAC Role

6.12 Mobile App

Objective: Give godown staff, delivery staff, field sales officers, and the owner a mobile-first way to operate and monitor the business, consistent with the product's ‘know your stock, sales, dues and performance — anytime, anywhere’ positioning.

Actors

- Godown Manager, Delivery Staff, Field Sales Officer, Owner

Functional Requirements

- **FR-MOB-01** Mobile app shall provide the Owner Dashboard (Section 6.9) with equivalent core figures to the web view.
- **FR-MOB-02** Mobile app shall support stock receipt/GRN entry and physical stock count capture by godown staff.
- **FR-MOB-03** Mobile app shall support proof-of-delivery capture (signature/photo/OTP) by delivery staff (FR-DEL-03).
- **FR-MOB-04** Mobile app shall support field sales order booking and scheme communication by field sales officers, even with intermittent rural connectivity (offline capture with sync-on-connectivity).
- **FR-MOB-05** Mobile app shall support barcode/QR scanning for product and batch identification in the godown.
- **FR-MOB-06** Mobile app shall support role-based views so a single app serves Owner, Godown, Delivery, and Field Sales personas with appropriate feature sets.

7. Data Model Summary

Core entities that anchor Mrida OS — Fertilizer Trading and how they relate. Full field-level schemas to be maintained in the engineering data-dictionary, not this PRD.

Entity	Key Relationships
Supplier	Parent of Purchase Order, Supplier Price List
Product / SKU	Parent of Price List, Stock Ledger, Sales/Purchase Invoice Line
Purchase Order / GRN	Links Supplier + Product; parent of Purchase Invoice, Batch/Lot Record
Batch / Lot	Child of GRN; carried through Stock Ledger to Sales Invoice Line for traceability
Godown	Parent of Stock Ledger entries; source/destination of Stock Transfer
Customer / Farmer	Parent of Sales Invoice, Outstanding Ledger, Credit Limit
Sales Invoice	Links Customer + Product/Batch + optional PoS Transaction Reference; parent of Delivery Challan
PoS Transaction Reference	Linked to Sales Invoice Line for subsidised fertilizer sales
Outstanding Ledger / Payment Receipt	Linked to Customer; updated by Sales Invoice and Payment Receipt
Delivery Challan	Child of Sales Invoice; parent of Proof of Delivery
Tally Voucher Sync Queue	Mirrors Purchase Invoice, Sales Invoice, and Payment Receipt for Tally push

8. Non-Functional Requirements

Category	Requirement
Performance	Billing and stock-check screens shall respond within 2 seconds under normal load; dashboard queries within 5 seconds.
Offline Support	Field sales order booking and godown stock-count capture must function offline and sync on reconnection without data loss.
Security & Access	Role-based access control at module and field level; encrypted storage of KYC/Aadhaar reference and bank details; audit logging on all financial and master-data changes.
Data Protection	Aadhaar/ID references stored per applicable data-protection requirements — masked in UI, encrypted at rest, access-logged.
Scalability	Architecture must support multi-branch, multi-godown concurrent operation and horizontal scaling as the anchor client's dealer network is onboarded.
Multi-tenancy readiness	Data model must be tenant-partitioned from day one, even if Phase 1 serves a single trading firm, to support replication across peer fertilizer dealers.
Localization	UI and SMS/WhatsApp templates must support English plus at least one regional language relevant to the anchor client's operating state.
Availability	Target 99.5% uptime, with particular resilience during peak sowing-season billing windows.
Data Retention	Sales, PoS transaction reference, and stock register records retained per statutory minimum retention period.

9. Integration Requirements

- Tally Prime — bidirectional voucher/ledger sync via XML-over-HTTP and ODBC (Section 6.10).
- Certified PoS device / DBT-in-Fertilizers ecosystem — transaction-reference mirroring for subsidised fertilizer sales (Section 6.8); direct iFMS/mFMS portal integration is out of Phase 1 scope (Section 5.2).
- GST e-invoicing / e-way bill — statutory sales/dispatch documentation above applicable thresholds.
- SMS / WhatsApp Business API — payment reminders, scheme broadcasts, delivery confirmations.
- Barcode/QR scanning — product and batch identification at godown and billing.
- Payment/banking — UPI/bank reconciliation for customer collections (manual reconciliation in Phase 1; gateway integration is a later-phase candidate per Section 5.2).

10. Phased Rollout Plan

Phase	Scope	Goal
Phase 1 — MVP	Supplier Mgmt, Purchase & Procurement, Inventory & Godown, Sales & Billing (incl. PoS routing), Owner Dashboard (single branch)	Prove the core purchase-to-sale-to-dashboard loop with the anchor client for one operating cycle
Phase 2 — Core Expansion	Customer & Farmer Mgmt, Credit & Payments, Delivery & Dispatch, Tally Integration, full Regulatory Compliance module	Cover the full order-to-cash cycle and remove duplicate bookkeeping against Tally
Phase 3 — Scale & Intelligence	Multi-branch/multi-godown, advanced BI/analytics, field sales mobile app, replication to peer fertilizer trading firms	Ready the product for horizontal licensing across the anchor client's dealer network

11. Assumptions & Dependencies

- Anchor client will provide real product, price list, and historical sales/purchase data for Phase 1 configuration and testing.
- Anchor client's certified PoS device vendor supports exposing a transaction reference/log that Mrida OS can mirror for reconciliation (Section 6.8); direct API access to the device is to be confirmed during Phase 1 discovery.
- Anchor client's Tally instance can be configured to run in server/ODBC-enabled mode on a network-reachable machine for integration (Section 6.10).
- Company will nominate module owners (Purchase, Sales, Accounts) to validate workflow detail during build, per the DIY-executable documentation approach.
- Regulatory parameters (subsidy purchase cap, MRP list, license renewal cadence) will be supplied/confirmed by the client and kept as configuration, not hard-coded, since they are subject to periodic government revision.

12. Risks & Mitigations

Risk	Mitigation
Counter staff resistance to an additional system alongside the PoS device and Tally	Phase 1 pilot with a small counter team; keep the additional data-entry burden minimal by auto-detecting subsidised items and pre-filling from Product Master

Risk	Mitigation
Tally sync failures going unnoticed until month-end close	Structural sync-status view (FR-TLY-03) with proactive alerts (FR-TLY-05), not a report staff must remember to check
Regulatory parameters (subsidy cap, MRP) changing without corresponding system update	Keep all regulatory figures as configuration values with a designated compliance owner responsible for updates, not hard-coded constants
Expiry-driven stock loss discovered too late	Structural Expiring Stock alerting (FR-INV-03) with configurable multi-stage lead time, not a manual register review
Multi-tenant architecture adds Phase 1 complexity for a single-client build	Tenant-partition data model from day one at low incremental cost, deferring true multi-tenant UI/billing to Phase 3

13. Glossary of Fertilizer Trading Terms

Term	Definition
FCO	Fertilizer Control Order — the statutory framework under which fertilizer manufacturing, marketing, and sale is licensed and regulated in India.
DBT (in Fertilizers)	Direct Benefit Transfer scheme under which the government pays fertilizer subsidy to manufacturers only against authenticated retail sales to farmers.
PoS (Point of Sale) Device	Certified device installed at a fertilizer retailer's shop used to authenticate the buyer (via Aadhaar/Voter ID/KCC) and record subsidised fertilizer sales.
iFMS	Integrated Fertilizer Management System — the central government platform capturing fertilizer sale transactions in real time for subsidy settlement and supply-chain monitoring.
mFMS	Mobile Fertilizer Management System — an earlier/complementary mobile-based mechanism for capturing retailer sale transactions.
MRP	Maximum Retail Price — for regulated fertilizer grades, a government-notified ceiling price that cannot be exceeded at sale.
FEFO	First-Expiry-First-Out — inventory issuing logic that dispatches the batch nearest to expiry first.
GRN	Goods Receipt Note — the record created when stock is physically received against a purchase order.
Voucher (Tally)	Tally's transaction record type (Sales, Purchase, Payment, Receipt, etc.) that Mrida OS must generate a compatible payload for during sync.

14. Open Questions for Stakeholder Input

The following need anchor-client confirmation before Phase 1 build starts, since they affect data model and workflow specifics:

- Which certified PoS device/vendor does the firm currently use for subsidised fertilizer billing, and does it expose any transaction log/API Mrida OS can read for reconciliation (Section 6.8)?
- Is the firm's Tally installation Tally Prime or Tally ERP 9, and can it be run in server/ODBC-enabled mode on a machine reachable from Mrida OS (Section 6.10)?
- How many godowns/branches does the firm operate today, and is multi-branch operation expected within the Phase 1–2 horizon?

- What proportion of sales is currently on credit versus cash, and what is the current method of tracking outstanding (register, Tally, informal)?
- Does the firm deal only in fertilizer, or also pesticides/seeds/other agri-inputs at meaningful volume — this affects how heavily the batch/expiry tracking (Section 6.3) needs to be prioritized in Phase 1?